

- DSM: Iterations, learning rate and batch size.
- Weibull AFT: Penalizer coefficient.

Tuning is done solely on the training bearings. We use the hyperparameters leading to the highest average concordance index (Antolini’s) on the validation folds to adjust the final models. Models are then evaluated on the test bearings.

Results

Table 2 shows the predictive performance of the survival models. We report the training time, Harrell’s concordance index (CI) (?), Antolini’s time-dependent concordance index (CI_{td}) (?) and the integrated Brier score (IBS) (?). Concerning the results on the XJTU dataset (see Tab. 2a), we observe that RSF, CoxBoost and DeepSurv perform the best in terms of ranking (CI) and DSM in terms of predictive accuracy. A CI of 0.70 means that 70 out of a 100 comparable pairs are ranked correctly. An IBS of 0.214 equals a 78.6% accuracy in predicting the survival function over all available times. Concerning the results on the PRONOSTIA dataset (see Tab. 2b), DSM, DeepSurv and WeibullAFT performs the best.

To compare our work to SOTA, (?) report a mean 70% accuracy (95% CI: 0.58-0.87) on eleven PRONOSTIA bearings. In our work, DeepSurv (?) performs the best in terms of predicting the survival function, yielding a 0.197 integrated Brier score on the test set. This translates to a predictive accuracy of 80.3%, however, the results are not directly comparable, as (?) use simple least-squares regression as opposed to survival regression, employ a different training/test split and only report the estimation error at the end of the bearing lifetime, instead of at available times.

Figure 4 shows inference results on the two XJTU test bearings. We see that all evaluated models provide consistent estimates of the survival function (leftmost panel). The Kaplan-Meier is used as a reference here and estimated on the full dataset (five bearings). In the central panel, we split the test data into two groups by the root mean square (RMS) covariate value, $RMS \leq 2$ and $RMS > 2$, and compute the mean survival curve of the two separately using a Cox proportional hazards model. This shows a clear separation in predicted survival probability, already from $t = 10$, to convergence at $t = 120$. The shadowed area represents a 95% confidence interval, which is wider for the bearings in the first group. The plot presents an interesting application for predictive maintenance, as it can be used as decision support for maintenance personnel. In the rightmost panel, by following (?), we computed 500 simulated survival times using a Cox model for the two RMS groups, and plotted their respective hazard function and survival time; the two hazard functions are quantitatively distinct, and the expected survival time for $RMS > 2$ is lower than for $RMS \leq 2$.

Conclusion

We have proposed survival analysis as a method to predict the risk of failure in ball bearings. First, we identify the point in time when a bearing starts to diverge significantly from its original frequency characteristic, which marks our event of interest, and second, we train several survival models to

estimate the risk of failure given a set of covariates sampled from the time-domain. We find the application of such methods interesting in manufacturing and engineering, as they inherently support censored data and provide a probabilistic prediction and confidence estimation instead of mere point estimates, thus we encourage further work within our framework.

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(a) XJTU ($N = 1000$, $C = 20\%$, $d = 12$).				
Model	T_{train}	CI $\uparrow$	CI _{td} $\uparrow$	IBS $\downarrow$
CoxPH	0.094	0.648	0.642	0.271
RSF	0.988	0.702	0.769	0.423
CoxBoost	2.791	0.59	0.769	0.261
DeepSurv	9.148	0.687	0.679	0.245
DSM	22.456	0.66	0.628	0.214
WeibullAFT	0.483	0.664	0.675	0.216

(b) PRONOSTIA ($N = 400$, $C = 20\%$, $d = 12$).				
Model	T_{train}	CI $\uparrow$	CI _{td} $\uparrow$	IBS $\downarrow$
CoxPH	0.016	0.626	0.625	0.34
RSF	0.265	0.715	0.701	0.383
CoxBoost	0.768	0.74	0.701	0.223
DeepSurv	2.392	0.745	0.705	0.197
DSM	9.737	0.739	0.748	0.294
WeibullAFT	0.489	0.764	0.691	0.236

Table 2: Performance metrics on the test sets. N : total sample size, C : pct. of censored data, d : number of covariates.

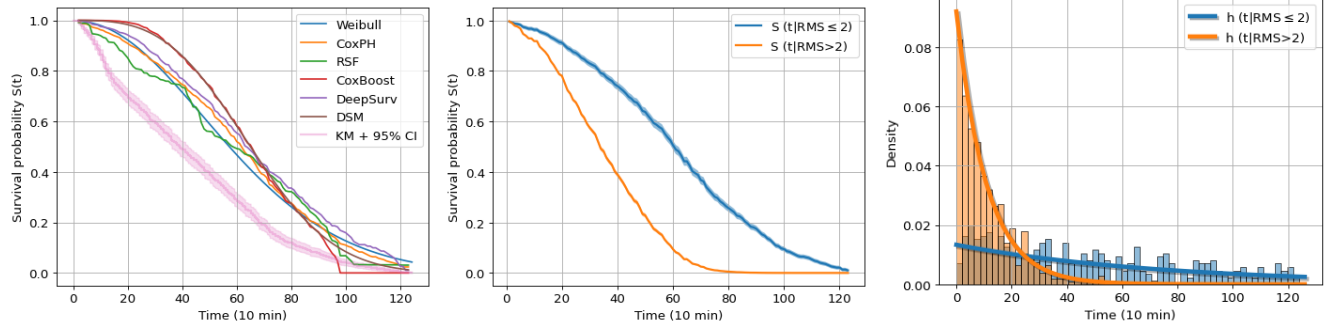


Figure 4: Model inference on the XJTU dataset. Left & center: survival function for different estimation approaches and for two separated groups based on RMS value (Solid lines: mean values; Shadow area: 95% conf. int.). Right: normalized survival times (bins) and hazard functions (solid lines) of the two RMS groups.